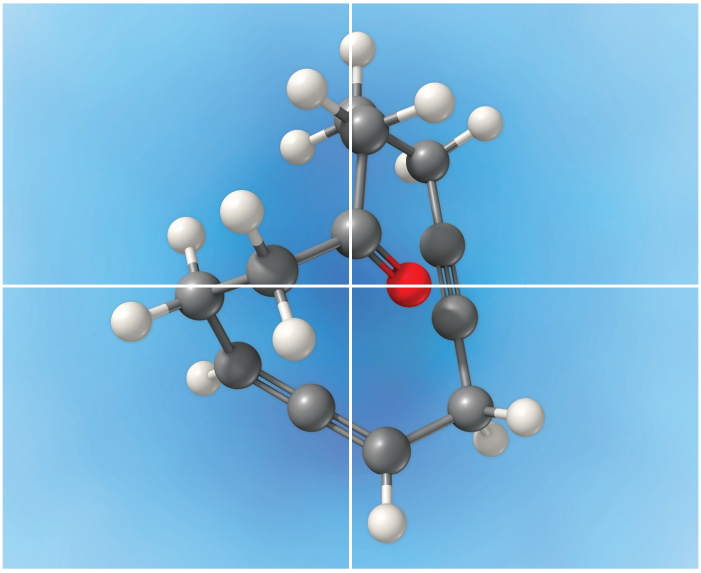
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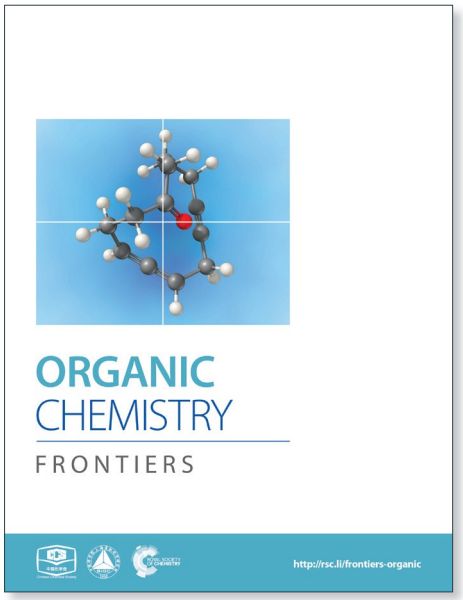


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ARTICLE

Controlled TfOH- or AuCl-catalyzed cycloisomerization and tandem hydrolytic defluorination of 1,2-allenyl perfluoroalkyl ketones

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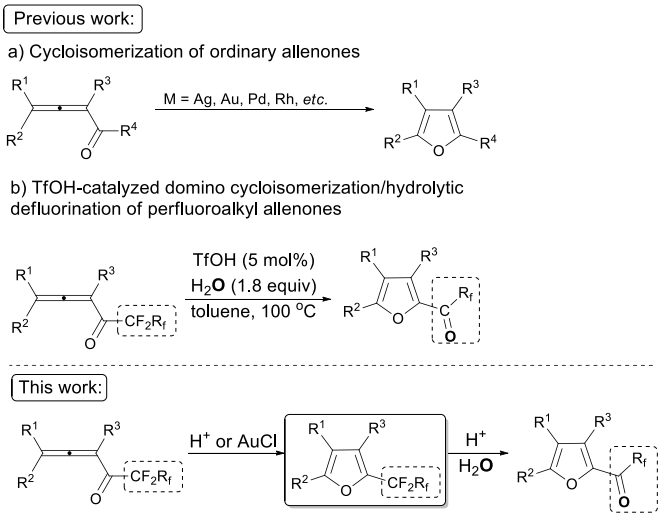
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Can Xue,[a] Xin Huang,[a] Shangze Wu,[a] Chunling Fu,*[a] and Shengming Ma*[a]

An efficient TfOH- or AuCl-catalyzed cycloisomerization reaction of easily available *n*-perfluoroalkyl 1,2-allenyl ketones occurring smoothly to form 2-perfluoroalkylfurans has been reported. In the presence of TfOH and H_2O, corresponding tandem hydrolytic defluorination afforded multi-substituted furan-2-yl *n*-perfluoroalkyl ketones.

Introduction

As one of the ubiquitous heterocyclic cores, furan plays a significant role in organic synthesis,[1] and also broadly exist in natural products[2] and useful materials.[3] Many efforts have been made to construct intricacy furan derivatives,[4-10] among which transition metal-catalyzed (such as Ag, Au, Pd, Rh, *etc.*) cycloisomerization of allenones[6-10] (a, Scheme 1) came out to be more apposite due to the substituent-loading capability of the unique substrates. On the other hand, due to the extensive application in the field of functional materials and the importance of organofluorine compounds,[11] the development of synthetic methodologies for generating perfluoroalkyl substituted heterocycles are of current interest. In our previous work, we have reported a TfOH-catalyzed[12] direct formation of furanyl perfluoroalkyl ketones from easily available *n*-perfluoroalkyl allenones[13] (b, Scheme 1).[14] In this study, we wish to describe the TfOH- or AuCl-catalyzed controlled cycloisomerization of perfluoroalkyl allenones affording perfluoroalkyl furans first, which may be converted to the corresponding hydrolytic defluorination ketones in the presence of TfOH and water.



Scheme 1. Cyclization reactions of ordinary allenones and perfluoroalkyl allenones.

Results and Discussion

We started to explore the cycloisomerization reaction with fully-substituted perfluoroalkyl allenone **1a** as the model substrate. No reaction occurred when AuCl was used as the catalyst (entry 1, Table 1), while $AuCl_3$ produced perfluoroalkyl furan **2a** in only 46% NMR yield (entry 2, Table 1). It is interesting for us to observe that aged triflic acid could catalyze the conversion of perfluoroalkyl substituted allenone **1a** to form furan **2a** together with furanyl ketone **3a** (entry 3, Table 1). To our delight, furan **2a** was formed in 96% NMR yield as the sole product (entry 5, Table 1) with a catalytic amount of newly purchased TfOH (5 mol%) in toluene at 60 °C for 10 h.

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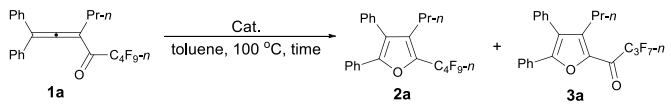
Electronic Supplementary Information (ESI) available: copies of ^{1}H NMR, ^{13}C NMR, ^{19}F NMR and NOE Spectra of all the compounds prepared. See DOI: 10.1039/x0xx00000x

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Table 1 Optimization of the cycloisomerization reaction of **1a**.[a]

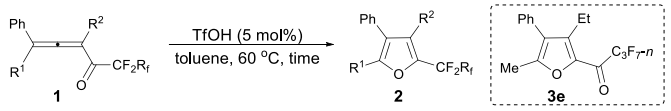


Entry	Cat. (mol%)	Time (h)	NMR Yield (%)[b]		Recovery of **1a** (%)[b]
			2a	**3a**	
1	AuCl (5)	10	0	0	82
2	$AuCl_3$ (10)	19	46	0	51
3	TfOH[c] (10)	2	35	59	0
4	TfOH[d] (10)	2.5	0	98	0
5[e]	TfOH (5)	10	96	0	0

[a] The reactions were conducted on 0.2 mmol scale in toluene (2 mL). [b] Determined by the ^{1}H NMR analysis with CH_2Br_2 as the internal standard. [c] Aged reagent, may contain water. [d] 1 equiv. H_2O was added with newly purchased TfOH. [e] The reaction was conducted at 60 °C and mesitylene was used instead of CH_2Br_2 as the internal standard.

The reaction scope was then explored following the optimized reaction conditions presented in entry 4 of Table 1. For the 1,1-diphenyl-substituted allenyl perfluoroalkyl ketones **1a-1c**, no matter R^3 = propyl or methyl, R_f = *n*-perfluoropropyl or pentyl, the corresponding **2a-2c** could be formed exclusively (entries 1-3, Table 2). Although **2d** could be obtained as the single product in good yield (entry 4, Table 2), changing R^1 from Et to Me, i.e., furan **2e** (69%) was formed together with ketone **3e**[14] (30%) (entry 5, Table 2).

Table 2 Cycloisomerization reactions of fully-substituted allenyl perfluoroalkyl ketones **1**.[a]

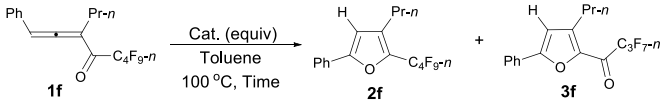


Entry	R^1	R^2	R_f	Time (h)	Yield of **2** (%)[b]
1[c]	Ph	*n*-Pr	*n*-C_3F_7 (**1a**)	10	96 (**2a**)
2[d]	Ph	*n*-Pr	*n*-C_5F_{11} (**1b**)	2	91 (**2b**)
3	Ph	Me	*n*-C_3F_7 (**1c**)	2.5	89 (**2c**)
4	Et	Et	*n*-C_3F_7 (**1d**)	4	75 (**2d**)
5	Me	Et	*n*-C_3F_7 (**1e**)	1.5	69 (**2e**)[e]

[a] Conditions: substrate (0.5 mmol), TfOH (5 mol%), toluene 5 mL, at 60 °C. [b] Isolated yield. [c] The reaction was conducted on 0.2 mmol scale. [d] TfOH (2 mol%) was used and the reaction was conducted on 0.3 mmol scale. [e] **3e** was also afforded in 30% isolated yield.

However, when we attempted to extend this cycloisomerization to 1-mono-substituted allenyl ketone **1f**, only 9% of furan **2f** was afforded with 87% recovery of **1f** (entry 1, Table 3). When trifluoromethanesulfonates such as $Cu(OTf)_2$[9f,9g] and AgOTf[9f,9g] were applied as the catalysts, there were still 88% and 56% of **1f** recovered, respectively (entries 2 and 3, Table 3). Interestingly, **2f** was obtained efficiently with a catalytic amount of AuCl in toluene at 100 °C (entry 4, Table 3). It should be mentioned that AuCl has been identified as the catalyst for the selective cycloisomerization of allenones for the first time.[9i] Then a series of other cheaper Lewis acids were screened: although 43% of **2f** could be produced when $FeCl_3$ was involved (entry 6, Table 3), 11% and 9% of **2f** were detected by NMR under the catalysis of $SnCl_4$ and $BF_3{\cdot}2H_2O$ (entries 7 and 8, Table 3), respectively; $BF_3{\cdot}Et_2O$ and CuI[9a] failed to catalyze the reaction (entries 9 and 10, Table 3); while with $TiCl_4$, complete decomposition of **1f** was obtained (entry 11, Table 4). Thus, the reaction conditions presented in entry 4 of Table 3 have been defined as the standard for further study.

Table 3 Optimization of the cycloisomerization reaction of **1f**.[a]



Entry	Cat. (equiv)	Time (h)	NMR yield (%)[b]		Recovery of **1f** (%)[b]
			2f	**3f**	
1[c]	TfOH (0.05)	11	9	2	87
2	$Cu(OTf)_2$ (0.1)	11	10	2	88
3	AgOTf (0.1)	11	42	2	56
4[d]	AuCl (0.05)	5	98 (90)	0	0
5[e]	$AuCl_3$ (0.05)	9	96	0	0
6	$FeCl_3$ (0.1)	24	43	0	55
7	$SnCl_4$ (1.0)	14	11	5	54
8	$BF_3{\cdot}2H_2O$ (3.0)	48	9	3	81
9	$BF_3{\cdot}Et_2O$ (3.0)	27	0	5	84
10	CuI (0.1)	19	0	0	88
11	$TiCl_4$ (1.0)	22	Complicated		0

[a] The reactions were conducted on 0.1 mmol scale in toluene (1 mL). [b] Determined by the ^{1}H NMR analysis with CH_2Br_2 as the internal standard and isolated yield was shown in the bracket. [c] The reaction was conducted at 60 °C. [d] The reaction was conducted on 0.5 mmol scale. [e] The reaction was conducted on 0.2 mmol scale.

With the optimized conditions in hand we then studied the generality of the AuCl-catalyzed cycloisomerization reactions of tri-substituted allenyl perfluorobutyl ketones **1f**-**1l**. The electronic effects of the substituent groups of the aryl group have almost no effect on the conversion (entries 4 and 5, Table 4). A heterocyclic 2-thienyl group and acyclic alkyl group could be compatible as well (entries 6 and 7, Table 4).

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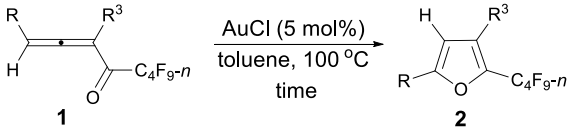
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Table 4 AuCl-catalyzed cycloisomerization of tri-substituted allenyl perfluorobutyl ketones **1**.[a]

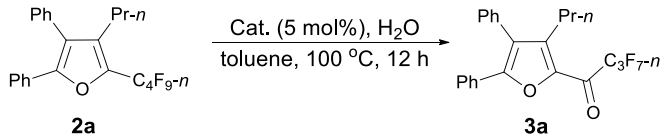


Entry	R	R^3	Time (h)	Yield of **2** (%)[b]
1	Ph	*n*-Pr (**1f**)	5	90 (**2f**)
2	Ph	Et (**1g**)	3	93 (**2g**)
3	Ph	Me (**1h**)	5	92 (**2h**)
4[c]	*p*-MeC_6H_4	*n*-Pr (**1i**)	12	98 (**2i**)
5	*p*-FC_6H_4	*n*-Pr (**1j**)	3	90 (**2j**)
6[c,d]	2-Thienyl	*n*-Pr (**1k**)	16	88 (**2k**)
7[c]	*n*-C_7H_{15}	*n*-Pr (**1l**)	11	56[e] (**2l**)

[a] Conditions: substrate (0.5 mmol), AuCl (5 mol%), toluene 5 mL, at 100 °C. [b] Isolated yield. [c] The reaction was carried out on a 0.2 mmol scale in toluene (2 mL). [d] AuCl (10 mol%) was used. [e] Purity (94.0%).

As shown in entry 3 of Table 1, beside normal cycloisomerization product furan **2a**, corresponding hydrolytic defluorination product ketone **3a**[14] could be obtained in 59% NMR yield as well. Thus, we attempted to form **3a** from **2a** in the presence of acid and water. To our delight, TfOH performed best among various acids. Moreover, 99% of **3a** could be generated with 1.5 equiv of water (entry 2, Table 5).

Table 5 Optimization of the hydrolytic defluorination reaction of **2a**.[a]

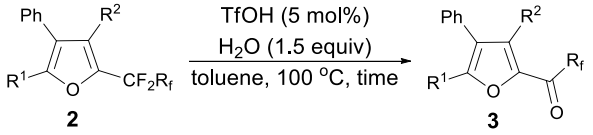


Entry	Cat.	H_2O (equiv)	NMR Yield of **3a** (%)[b]	Recovery of **2a** (%)[b]
1	TfOH	1.0	97	0
2[c]	TfOH	1.5	99	0
3	TfOH	2.0	98	0
4	TfOH	5.0	94	0
5[d]	TFA	1.5	0	94
6[e]	TsOH·H_2O	1.5	39	61

[a] The reactions were conducted on 0.1 mmol scale in toluene (1 mL) under a N_2 atmosphere. [b] Determined by the ^1H NMR analysis with CH_2Br_2 as the internal standard. [c] The reactions were conducted on 0.2 mmol scale in toluene (2 mL) under a N_2 atmosphere. [d] The dosage of catalyst is 10 mol%. [e] The dosage of catalyst is 1.0 equiv.

Following the optimized reaction conditions presented in entry 2 of Table 5, we studied the scope of this hydrolytic defluorination conversion. When R^1 = Ph, R^2 = acyclic alkyl, and R_f = *n*-C_3F_7 or *n*-C_5F_{11}, ketone product **3a-3c** could all be obtained with perfect yields (entries 1-3, Table 6). R^1 and R^2 may both be alkyl groups: **3e** was produced in 71% isolated yield (entry 4, Table 6).

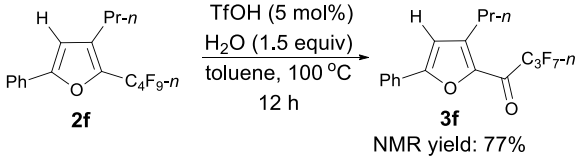
Table 6 Hydrolytic defluorination reactions of perfluoroalkyl furans **2**.[a]



Entry	R^1	R^2	R_f	Time (h)	Yield of **3** (%)[b]
1	Ph	*n*-Pr	*n*-C_3F_7 (**2a**)	12	97 (**3a**)
2	Ph	*n*-Pr	*n*-C_5F_{11} (**2b**)	10	95 (**3b**)
3	Ph	Me	*n*-C_3F_7 (**2c**)	10	95 (**3c**)
4	Me	Et	*n*-C_3F_7 (**2e**)	3	71 (**3e**)

[a] Conditions: substrate (0.2 mmol), TfOH (5 mol%), H_2O (1.5 equiv), in toluene 2 mL, at 100 °C. [b] Isolated yield.

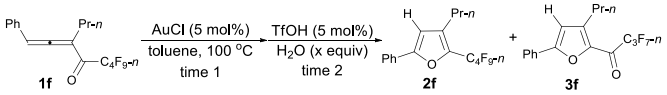
Furthermore, when tri-substituted furan **2f** was treated with TfOH (5 mol%) and H_2O (1.5 equiv), corresponding ketone **3f** could also be formed in good yield (eq.1).



eq. 1

Considering that **2f** could be efficiently prepared from **1f** with AuCl as the catalyst (entry 4, Table 3) and the acid-catalyzed reaction of **2f** would form **3f** (eq. 1), thus, a two-step process has been established (Table 7): the addition of 1.5 equiv of H_2O to the reactor vessel afforded the highest yield of pure **3f**, exclusively (entry 2, Table 7).

Table 7 Optimization of the dosage of water in the two-step one-pot process for the tandem cycloisomerization/hydrolytic defluorination reaction of **1f**.[a]



Entry	H_2O (equiv)	Time1/Time 2 (h)	NMR Yield (%)[b]	
			2f	**3f**
1	1	3.5/24	4	63
2	1.5	12/8	0	84
3	2	3.5/3	0	79
4	3	12/8	6	61
5	5	4/5	7	75
6[c]	10	4/5	Trace	50

[a] The reactions were conducted on 0.1 mmol scale in toluene (1.0 mL). [b] Determined by the ^1H NMR analysis with CH_2Br_2 as the internal standard. [c] A bit complicated.

With a reliable procedure in hand, we next turned our attention to study the scope of the two-step one-pot transformation: when Ar = Ph, the tandem reaction could all proceed efficiently no matter R^1 was a propyl, an ethyl, or a methyl group (entries 1-4, Table 8); the substituent of the aryl group has almost no significant effect on the reaction (entries 5 and 6, Table 8); albeit with a lower yield 2-thienyl group may

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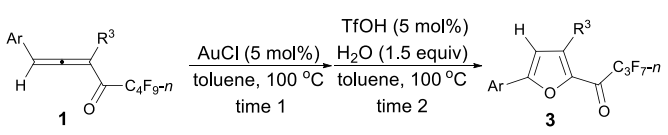
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also be compatible (entry 7, Table 8). To demonstrate the practicality of the procedure, a gram scale (2.50 mmol) reaction of **1f** was also conducted in good yield (78%) as shown in entry 2, Table 8.

Conclusions

In conclusion, we have developed an efficient controlled cycloisomerization of allenyl perfluoroalkyl ketones under the catalysis of AuCl (for 3-monoaryl allenones) or TfOH (for fully-substituted allenones) to afford 2-perfluoroalkyl furans. In the presence of water and a catalytic amount of TfOH, these furans would be efficiently transformed into corresponding furan-2-yl perfluoroalkyl ketones. Further studies in this area are still undergoing in our laboratory.

Table 8 The two-step one-pot tandem cycloisomerization/hydrolytic defluorination reaction of 3-aryl-substituted allenyl perfluorobutyl ketones **1**.[a]



Entry	Ar	R^1	Time 1/Time 2 (h)	Yield of **3** (%)[b]
1[c]	Ph	*n*-Pr (**1f**)	3/3	76 (**3f**)
2[d]	Ph	*n*-Pr (**1f**)	3/3.5	78 (**3f**)
3	Ph	Et (**1g**)	13/10	86 (**3g**)
4[c]	Ph	Me (**1h**)	3.5/13	86 (**3h**)
5	*p*-MeC_6H_4	*n*-Pr (**1i**)	11/5	88 (**3i**)
6	*p*-FC_6H_4	*n*-Pr (**1j**)	12/10	82 (**3j**)
7[e]	2-Thienyl	*n*-Pr (**1k**)	13/10	43 (**3k**)

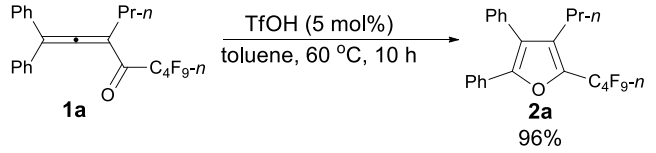
[a] Conditions: substrate (0.5 mmol), AuCl (5 mol%), toluene 5 mL, at 100 °C for time 1, then TfOH (5 mol%) and H_2O (1.5 equiv) were added. [b] Isolated yield. [c] The reaction was conducted on a 1.0 mmol scale. [d] The reaction was carried out on a gram scale. [e] The reaction was carried out on a 0.2 mmol scale.

Experimental Section

General： Perfluoroalkyl allenones were prepared according to the literature procedures.[13] Toluene was distilled from Na wire/ benzophenone. TfOH (Alfa) was stored in a glovebox, and transferred with a microsyringe. Other commercially available chemicals were purchased and used without additional purification. The reactions were performed under an atmosphere of nitrogen using standard Schlenk tubes unless otherwise. Petroleum ether with a boiling point ranges from 30 to 60 °C was used. Flash-column chromatography was carried out on silica gel H (10–40 μ). ^{1}H NMR spectra (300 MHz) were recorded using TMS as an internal standard (δ 0 ppm). ^{13}C NMR spectra (75 MHz) were recorded using $CDCl_3$ as an internal standard (δ 77.00 ppm). ^{19}F NMR spectra (282 MHz) were recorded using $CFCl_3$ as an internal standard (δ 0 ppm). IR spectra were recorded with a Perkin-Elmer 983G instrument. Mass spectrometry was taken with an HP 5989A system. High-resolution mass spectrometry was determined with a Finnigan MAT 8430 or Bruker APEXIII instrument. The structure of **2d**, **2e**, **2g**, **2k**, and **2l** were established by the NOE study.

1. Typical Procedure I: TfOH-catalyzed cycloisomerization reactions of fully-substituted allenyl perfluoroalkyl ketones **2a-2e**.

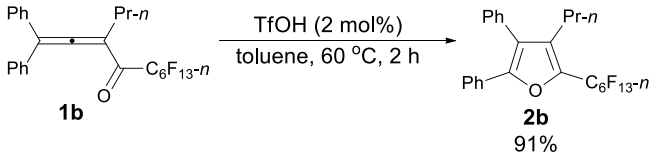
(1) Synthesis of 4,5-diphenyl-2-(*n*-perfluorobutyl)-3-(*n*-propyl)furan (**2a**)



Typical Procedure I: To a dried Schlenk tube were added 1,1-diphenylhexa-1,2-dien-3-yl perfluorobutyl ketone **1a** (0.0961 g, 0.20 mmol)/rinsed with toluene (2 mL) and TfOH (0.9 μL, *d* = 1.695 g/mL, 0.0015 g, 0.01 mmol) at room temperature under a nitrogen atmosphere. The resulting reaction mixture was placed in a pre-heated oil bath of 60 °C and strirried for 10 h as monitored by TLC. After being cooled to the room temperature, the crude reaction mixture was transferred to a round bottom flask. After removing the solvent via rotary evaporation, column chromatography on silica gel (eluent: petroleum ether) afforded **3a** (0.0924 g, 96%) as a waxy solid: ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.49-7.33 (m, 5 H, ArH), 7.33-7.26 (m, 2 H, ArH), 7.25-7.18 (m, 3 H, ArH), 2.42 (t, *J* = 8.0 Hz, 2 H, CH_2), 1.42-1.28 (m, 2 H, CH_2), 0.78 (t, *J* = 7.4 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 151.2, 134.4 (t, *J* = 31.7 Hz), 132.54, 132.49, 130.1, 129.7, 129.0, 128.4, 128.3, 128.1, 126.0, 124.4, 25.0, 23.6, 13.9; ^{19}F NMR (282 MHz, $CDCl_3$) δ -81.4 (t, *J* = 9.7 Hz, 3 F), -110.5~-110.8 (m, 2 F), -123.6~-123.8 (m, 2 F), -126.5~-126.7 (m, 2 F); IR (neat) ν (cm^{-1}) 3078, 3029, 2981, 2951, 2917, 1642, 1610, 1494, 1448, 1432, 1405, 1315, 1294, 1268, 1213, 1157, 1132, 1095, 1057; MS (70 ev, EI) *m/z* (%) 480 (M^+, 50.59), 311 (100); HRMS calcd for $C_{23}H_{17}F_9O$ (M^+): 480.1136, Found: 480.1124.

The following compounds **2b-2e** were prepared following **Typical Procedure I**.

(2) Synthesis of 4,5-diphenyl-2-(*n*-perfluorohexyl)-3-(*n*-propyl)furan (**2b**)



The reaction of 1,1-diphenylhexa-1,2-dien-3-yl *n*-perfluorohexyl ketone **1b** (0.1735 g, 0.30 mmol) with TfOH (99%, 0.54 μL, *d* = 1.695 g/mL, 0.0009 g, 0.006 mmol) in toluene (3 mL) at 60 °C for 2 h afforded **2b** (0.1574 g, 91%) (petroleum ether) as a solid: m.p. 53.4-55.2 °C (*n*-hexane/ethyl acetate); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.53-7.37 (m, 5 H, ArH), 7.37-7.30 (m, 2 H, ArH), 7.29-7.10 (m, 3 H, ArH), 2.45 (tt, J_1 = 7.8 Hz, J_2 = 1.8 Hz, 2 H, CH_2), 1.46-1.30 (m, 2 H, CH_2), 0.81 (t, *J* =

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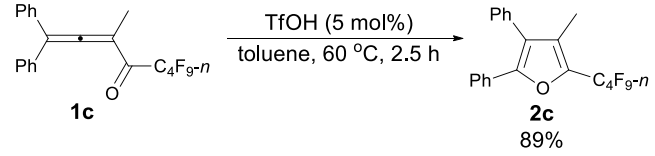
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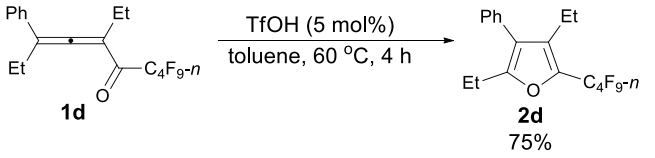
7.4 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 151.2, 134.5 (t, *J* = 32.2 Hz), 132.6, 132.5, 130.1, 129.7, 129.0, 128.4, 128.3, 128.1, 126.0, 124.4, 25.1, 23.6, 13.8; ^{19}F NMR (282 MHz, $CDCl_3$) δ -81.3 (t, *J* = 9.7 Hz, 3 F), -110.5 (t, *J* = 12.3 Hz, 2 F), -122.2~-122.6 (m, 2 F), -122.6~-122.9 (m, 2 F), -123.0~-123.5 (m, 2 F), -126.5~-126.8 (m, 2 F); IR (KBr) ν (cm^{-1}) 3064, 3033, 2967, 2937, 2877, 1623, 1605, 1583, 1561, 1505, 1482, 1467, 1447, 1402, 1382, 1362, 1333, 1239, 1095, 1073, 1057, 1029; MS (70 ev, EI) *m/z* (%) 581 (31.20), 580 (M^+, 100); Elemental analysis calcd for $C_{25}H_{17}F_{13}O$: C, 51.74; H, 2.95; Found: C, 51.74; H, 3.12.

(3) Synthesis of 4,5-diphenyl-3-methyl-2-(*n*-perfluorobutyl)furan (**2c**)



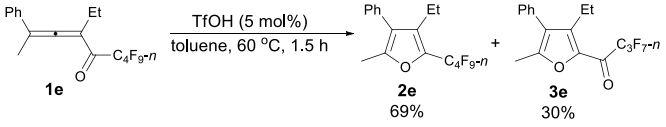
The reaction of 1,1-diphenylbuta-1,2-dien-3-yl *n*-perfluorobutyl ketone **1c** (0.2265 g, 0.50 mmol) with TfOH (99%, 2.3 μL, *d* = 1.695 g/mL, 0.0039 g, 0.026 mmol) in toluene (5 mL) at 60 °C for 2.5 h afforded **2c** (0.2020 g, 89%) (petroleum ether/ethyl acetate = 100/1) as a solid: m.p. 52.3-53.2 °C (*n*-hexane/ethyl acetate); ^1H NMR (300 MHz, $CDCl_3$) δ 7.58-7.02 (m, 10 H, ArH), 2.03 (s, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 151.0, 134.5 (t, *J* = 31.8 Hz), 132.1, 130.1, 129.7, 129.0, 128.4, 128.3, 128.0, 127.6, 126.1, 124.7, 8.5; ^{19}F NMR (282 MHz, $CDCl_3$) δ -81.5 (t, *J* = 9.7 Hz, 3 F), -111.2~-111.6 (m, 2 F), -123.9~-124.4 (m, 2 F), -126.5~-126.9 (m, 2 F); IR (KBr) ν (cm^{-1}) 3064, 3033, 2936, 1629, 1605, 1584, 1563, 1505, 1481, 1447, 1402, 1351, 1335, 1235, 1203, 1135, 1110, 1081, 1030, 1006; MS (70 ev, EI) *m/z* (%) 452 (M^+, 48.75), 283 (100); Elemental analysis calcd for $C_{21}H_{13}F_9O$: C, 55.76; H, 2.90; Found: C, 55.96; H, 3.19.

(4) Synthesis of 3,5-diethyl-4-phenyl-2-(*n*-perfluorobutyl)furan (**2d**)



The reaction of 3-phenylhepta-3,4-dien-5-yl *n*-perfluorobutyl ketone **1d** (0.2090 g, 0.50 mmol) with TfOH (99%, 2.3 μL, *d* = 1.695 g/mL, 0.0039 g, 0.026 mmol) in toluene (5 mL) at 60 °C for 4 h afforded **2d** (0.1569 g, 75%) (petroleum ether) as a liquid: ^1H NMR (300 MHz, $CDCl_3$) δ 7.50-7.31 (m, 3 H, ArH), 7.31-7.20 (m, 2 H, ArH), 2.59 (q, *J* = 7.5 Hz, 2 H, CH_2), 2.50 (q, *J* = 7.4 Hz, 2 H, CH_2), 1.18 (t, *J* = 7.5 Hz, 3 H, CH_3), 0.93 (t, *J* = 7.5 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 156.6, 133.1 (t, *J* = 32.0 Hz), 132.3, 132.2, 129.7, 128.6, 127.5, 123.2, 19.9, 16.5, 15.0, 12.6; ^{19}F NMR (282 MHz, $CDCl_3$) δ -81.5 (t, *J* = 10.4 Hz, 3 F), -111.0 (t, *J* = 10.6 Hz, 2 F), -123.9~-124.3 (m, 2 F), -126.6~-127.0 (m, 2 F); IR (neat) ν (cm^{-1}) 3035, 2979, 2941, 2883, 1630, 1576, 1496, 1466, 1408, 1378, 1353, 1324, 1276, 1236, 1135, 1091, 1062, 1004; MS (70 ev, EI) *m/z* (%) 418 (M^+, 66.28), 249 (100); HRMS calcd for $C_{18}H_{15}F_9O$ (M^+): 418.0979, Found: 418.0977.

(5) Synthesis of 3-ethyl-5-methyl-2-(*n*-perfluorobutyl)-4-phenylfuran (**2e**) and (3-ethyl-5-methyl-4-phenylfuran-2-yl) *n*-perfluoropropyl ketone (**3e**)



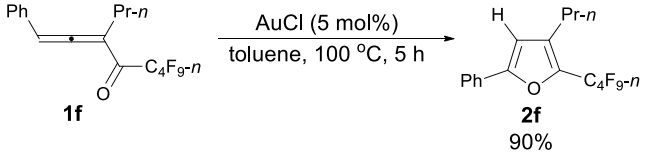
The reaction of 2-phenylhexa-2,3-dien-4-yl perfluorobutyl ketone **1e** (0.2025 g, 0.50 mmol) with TfOH (99%, 2.3 μL, *d* = 1.695 g/mL, 0.0039 g, 0.026 mmol) in toluene (5 mL) at 60 °C for 1.5 h afforded **2e** (0.1389 g, 69%) and **3e** (0.0567 g, 30%) (petroleum ether, then petroleum ether/ethyl acetate = 200/1).

2e (liquid, less polar): ^1H NMR (300 MHz, $CDCl_3$) δ 7.49-7.32 (m, 3 H, ArH), 7.29-7.20 (m, 2 H, ArH), 2.51 (q, *J* = 7.5 Hz, 2 H, CH_2), 2.26 (s, 3 H, CH_3), 0.93 (t, *J* = 7.5 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 151.7, 133.2 (t, *J* = 31.8 Hz), 132.4, 132.2, 129.6, 128.6, 127.5, 123.9, 16.5, 15.0, 12.1; ^{19}F NMR (282 MHz, $CDCl_3$) δ -81.5 (t, *J* = 9.7 Hz, 3 F), -110.9 (t, *J* = 10.7 Hz, 2 F), -123.7~-124.1 (m, 2 F), -126.6~-126.9 (m, 2 F); IR (neat) ν (cm^{-1}) 3084, 3061, 3036, 2978, 2941, 2883, 1633, 1578, 1497, 1467, 1445, 1411, 1354, 1279, 1235, 1135, 1091, 1055, 1012; MS (70 ev, EI) *m/z* (%) 404 (M^+, 52.43), 235 (100); HRMS calcd for $C_{17}H_{13}F_9O$ (M^+): 404.0823, Found: 404.0815.

3e (liquid, more polar): ^1H NMR (300 MHz, $CDCl_3$) δ 7.51-7.33 (m, 3 H, ArH), 7.29-7.21 (m, 2 H, ArH), 2.78 (q, *J* = 7.5 Hz, 2 H, CH_2), 2.38 (s, 3 H, CH_3), 1.10 (t, *J* = 7.4 Hz, 3 H, CH_3); ^{19}F NMR (282 MHz, $CDCl_3$) δ -81.0 (t, *J* = 9.0 Hz, 3 F), -117.6~-118.1 (m, 2 F), -126.2~-126.6 (m, 2 F).

2. Typical Procedure II: AuCl-catalyzed cycloisomerization of 3-monoaryl-substituted allenyl perfluorobutyl ketones **1f-1l**.

(1) Synthesis of 2-(*n*-perfluorobutyl)-5-phenyl-3-(*n*-propyl)furan (**2f**)



Typical Procedure II: To a dried Schlenk tube were added anhydrous AuCl (97%, 0.0062 g, 0.026 mmol), 1-phenylhexa-1,2-dien-3-yl *n*-perfluorobutyl ketone **1f** (0.2022 g, 0.50 mmol), and anhydrous toluene (5 mL) at room temperature under N_2 atmosphere. The resulting mixture was then placed in a pre-heated oil bath of 100 °C and stirred for 5 h as monitored by TLC. After being cooled to room temperature, the crude reaction mixture was filtrated through a short column of silica gel eluted with 60 mL of Et_2O. After removing the solvent via rotary evaporation, column chromatography on silica gel (eluent: petroleum ether) afforded **2f** (0.1818 g, 90%) as an oil: ^1H NMR (300 MHz, $CDCl_3$) δ 7.73-7.63 (m, 2 H, ArH), 7.46-7.27 (m, 3 H, ArH), 6.62 (s, 1 H, =CH), 2.56 (tt, J_1 = 7.5 Hz, J_2 = 1.8 Hz,

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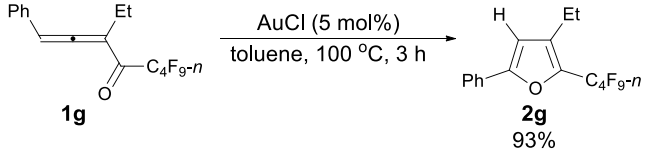
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2 H, CH_2), 1.74-1.56 (m, 2 H, CH_2), 0.98 (t, *J*= 7.4 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 155.9, 134.5 (t, *J* = 32.3 Hz), 132.6, 129.4, 128.8, 128.7, 124.4, 107.9, 26.3, 23.2, 13.7; ^{19}F NMR (282 MHz, $CDCl_3$) δ -81.5 (t, *J* = 10.0 Hz, 3 F), -110.7 (t, *J* = 10.6 Hz, 2 F), -123.9~-124.3 (m, 2 F), -126.5~-126.9 (m, 2 F); IR (neat) ν (cm^{-1}) 3066, 3038, 2965, 2933, 2876, 1948, 1877, 1802, 1750, 1621, 1605, 1584, 1551, 1487, 1451, 1397, 1350, 1313, 1235, 1135, 1095, 1057; MS (70 ev, EI) *m/z* (%) 404 (M^+, 30.78), 235 (100); HRMS calcd for $C_{17}H_{13}F_9O$ (M^+): 404.0823, Found: 404.0836.

The following compounds **2g-2l** were prepared following **Typical Procedure II**.

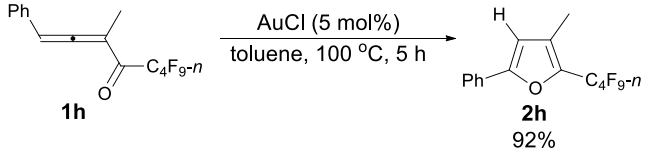
(2) Synthesis of 3-ethyl-2-(*n*-perfluorobutyl)-5-phenylfuran (**2g**)



1g **2g** 93%

The reaction of 1-phenylpenta-1,2-dien-3-yl *n*-perfluorobutyl ketone **1g** (0.1951 g, 0.50 mmol) with AuCl (99%, 0.0056 g, 0.024 mmol) in toluene (5 mL) at 100 °C for 3 h afforded **2g** (0.1808 g, 93%) (petroleum ether) as a liquid: ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.73-7.61 (m, 2 H, ArH), 7.46-7.27 (m, 3 H, ArH), 6.64 (s, 1 H, =CH), 2.71-2.50 (m, 2 H, CH_2), 1.23 (t, *J* = 7.7 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 156.0, 134.2, 134.0 (t, *J* = 31.7 Hz), 129.5, 128.8, 128.7, 124.4, 107.6, 17.8, 14.5; ^{19}F NMR (282 MHz, $CDCl_3$) δ -81.5 (t, *J* = 9.9 Hz, 3 F), -111.1 (t, *J* = 10.6 Hz, 2 F), -124.1~-124.4 (m, 2 F), -126.6~-127.1 (m, 2 F); IR (neat) ν (cm^{-1}) 3067, 3039, 2980, 2942, 2884, 1621, 1605, 1585, 1551, 1487, 1466, 1452, 1398, 1380, 1351, 1282, 1235, 1135, 1091, 1003; MS (70 ev, EI) *m/z* (%) 390 (M^+, 37.26), 221 (100); HRMS calcd for $C_{16}H_{11}F_9O$ (M^+): 390.0666, Found: 390.0664.

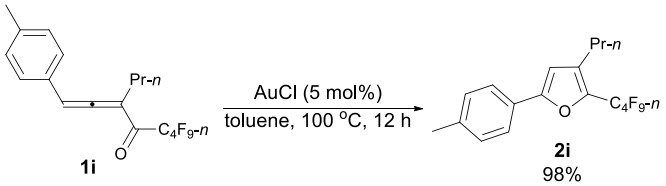
(3) Synthesis of 3-methyl-2-(*n*-perfluorobutyl)-5-phenylfuran (**2h**)



1h **2h** 92%

The reaction of 1-phenylbuta-1,2-dien-3-yl *n*-perfluorobutyl ketone **1h** (0.1891 g, 0.50 mmol) with AuCl (97%, 0.0062 g, 0.026 mmol) in toluene (5 mL) at 100 °C for 5 h afforded **2h** (0.1749 g, 92%) (petroleum ether) as a liquid: ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.74-7.60 (m, 2 H, ArH), 7.46-7.26 (m, 3 H, ArH), 6.56 (s, 1 H, =CH), 2.21 (s, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 155.8, 134.8 (t, *J* = 31.9 Hz), 129.4, 128.8, 128.7, 127.6, 124.4, 109.4, 10.0; ^{19}F NMR (282 MHz, $CDCl_3$) δ -81.5 (t, *J* = 9.4 Hz, 3 F), -111.2~-111.6 (m, 2 F), -124.2~-124.6 (m, 2 F), -126.6~-127.1 (m, 2 F); IR (neat) ν (cm^{-1}) 3067, 3039, 2936, 1949, 1878, 1802, 1751, 1626, 1606, 1585, 1551, 1488, 1451, 1401, 1384, 1350, 1235, 1135, 1087, 1056, 1027, 1005; MS (70 ev, EI) *m/z* (%) 376 (M^+, 47.65), 207 (100); HRMS calcd for $C_{15}H_9F_9O$ (M^+): 376.0510, Found: 376.0515.

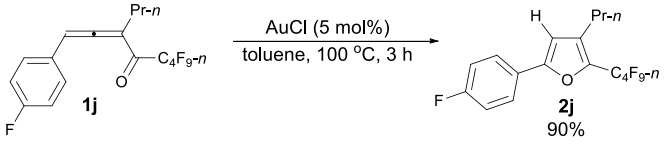
(4) Synthesis of 5-(4'-methylphenyl)-2-(*n*-perfluorobutyl)-3-(*n*-propyl)furan (**2i**)



1i **2i** 98%

The reaction of 1-(4'-methylphenyl)hexa-1,2-dien-3-yl perfluorobutyl ketone **1i** (0.0832 g, 0.20 mmol) with AuCl (97%, 0.0023 g, 0.010 mmol) in toluene (2 mL) at 100 °C for 12 h afforded **2i** (0.0812 g, 98%) (eluent: petroleum ether) as a liquid: ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.56 (d, *J* = 8.1 Hz, 2 H, ArH), 7.20 (d, *J* = 8.1 Hz, 2 H, ArH), 6.56 (s, 1 H, =CH), 2.55 (t, *J* = 7.7 Hz, 2 H, CH_2), 2.37 (s, 3 H, CH_3), 1.74-1.54 (m, 2 H, CH_2), 0.97 (t, *J* = 7.4 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 156.1, 138.7, 134.0 (t, *J* = 32.3 Hz), 132.6, 129.5, 126.7, 124.3, 107.2, 26.3, 23.2, 21.3, 13.7; ^{19}F NMR (282 MHz, $CDCl_3$) δ -81.5 (t, *J* = 9.4 Hz, 3 F), -110.8 (t, *J* = 11.6 Hz, 2 F), -124.0~-124.3 (m, 2 F), -126.6~-126.9 (m, 2 F); IR (neat) ν (cm^{-1}) 3029, 2967, 2935, 2876, 1906, 1621, 1581, 1557, 1498, 1468, 1417, 1393, 1350, 1315, 1235, 1135, 1094, 1053; MS (70 ev, EI) *m/z* (%) 418 (M^+, 20.78), 249 (100); HRMS calcd for $C_{18}H_{15}F_9O$ (M^+): 418.0979, Found: 418.0978.

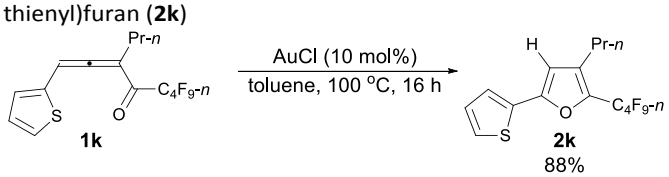
(5) Synthesis of 5-(4'-fluorophenyl)-2-(*n*-perfluorobutyl)-3-(*n*-propyl)furan (**2j**)



1j **2j** 90%

The reaction of 1-(4'-fluorophenyl)hexa-1,2-dien-3-yl *n*-perfluorobutyl ketone **1j** (0.2120 g, 0.50 mmol) with AuCl (97%, 0.0061 g, 0.026 mmol) in toluene (5 mL) at 100 °C for 3 h afforded **2j** (0.1909 g, 90%) (petroleum ether) as a liquid: ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.63 (dd, J_1 = 8.4 Hz, J_2 = 5.4 Hz, 2 H, ArH), 7.08 (t, *J* = 8.7 Hz, 2 H, ArH), 6.54 (s, 1 H, =CH), 2.55 (t, *J* = 7.5 Hz, 2 H, CH_2), 1.72-1.56 (m, 2 H, CH_2), 0.98 (t, *J* = 7.4 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 163.0 (d, *J* = 247.4 Hz), 155.1, 134.6 (t, *J* = 31.6 Hz), 132.8, 126.3 (d, *J* = 8.1 Hz), 125.9 (d, *J* = 2.3 Hz), 115.9 (d, *J* = 21.9 Hz), 107.6, 26.3, 23.2, 13.6; ^{19}F NMR (282 MHz, $CDCl_3$) δ -81.4~-81.6 (m, 3 F), -110.6~-111.0 (m, 2 F), -112.4~-112.7 (m, 1 F), -123.9~-124.3 (m, 2 F), -126.5~-126.9 (m, 2 F); IR (neat) ν (cm^{-1}) 3052, 2969, 2937, 2878, 1891, 1621, 1607, 1559, 1496, 1469, 1421, 1392, 1351, 1235, 1135, 1097, 1054, 1014; MS (70 ev, EI) *m/z* (%) 422 (M^+, 65.02), 253 (100); HRMS calcd for $C_{17}H_{12}F_{10}O$ (M^+): 422.0728, Found: 422.0730.

(6) Synthesis of 2-(*n*-perfluorobutyl)-3-(*n*-propyl)-5-(2-thienyl)furan (**2k**)



1k **2k** 88%

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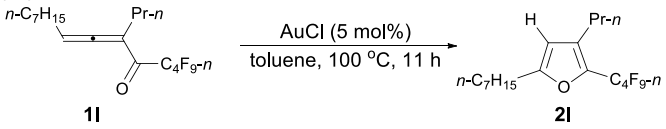
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The reaction of 1-(2-thienyl)hexa-1,2-dien-3-yl *n*-perfluorobutyl ketone **1k** (0.0828 g, 0.20 mmol) with AuCl (97%, 0.0049 g, 0.020 mmol) in toluene (2 mL) at 100 °C for 16 h afforded **2k** (0.0727 g, 88%) (petroleum ether) as a liquid: ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.38-7.26 (m, 2 H, ArH), 7.05 (dd, J_1 = 4.8 Hz, J_2 = 3.9 Hz, 1 H, ArH), 6.46 (s, 1 H, =CH), 2.53 (t, *J* = 7.5 Hz, 2 H, CH_2), 1.71-1.55 (m, 2 H, CH_2), 0.97 (t, *J* = 7.4 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 151.5, 134.0 (t, *J* = 32.0 Hz), 132.7, 132.1, 127.8, 125.8, 124.5, 107.8, 26.3, 23.2, 13.6; ^{19}F NMR (282 MHz, $CDCl_3$) δ -81.5 (t, *J* = 10.2 Hz, 3 F), -110.7 (t, *J* = 11.0 Hz, 2 F), -123.9~-124.2 (m, 2 F), -126.6~-126.9 (m, 2 F); IR (neat) ν (cm^{-1}) 3110, 3081, 2967, 2935, 2877, 1792, 1620, 1575, 1495, 1468, 1424, 1400, 1350, 1235, 1135, 1094, 1051, 1019; MS (70 ev, EI) *m/z* (%) 410 (M^+, 32.52), 241 (100); HRMS calcd for $C_{15}H_{11}F_9OS$ (M^+): 410.0387, Found: 410.0387.

(7) Synthesis of 2-(*n*-perfluorobutyl)-5-heptyl-3-(*n*-propyl)furan (**2l**)

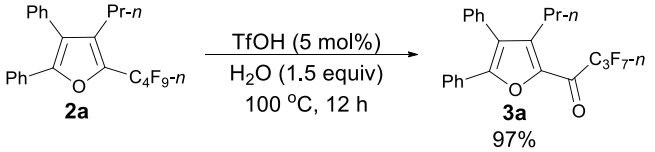


1l **2l** 56%

The reaction of trideca-4,5-dien-4-yl *n*-perfluorobutyl ketone **1l** (0.0854 g, 0.20 mmol) with AuCl (99%, 0.0023 g, 0.01 mmol) in toluene (2 mL) at 100 °C for 11 h afforded **2l** (0.0513 g, purity 94.0% as determined by using CH_2Br_2 as the internal standard, 56% yield) (petroleum ether) as an oil: ^{1}H NMR (300 MHz, $CDCl_3$) δ 5.98 (s, 1 H, =CH), 2.59 (t, *J* = 7.7 Hz, 2 H, CH_2), 2.52-2.42 (m, 2 H, CH_2), 1.68-1.50 (m, 4 H, 2 × CH_2), 1.35-1.23 (m, 8 H, 4 × CH_2), 0.93 (t, *J* = 7.4 Hz, 3 H, CH_3), 0.88 (t, *J* = 6.8 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 159.4, 133.2 (t, *J* = 31.9 Hz), 131.5, 108.5, 31.7, 29.0, 28.9, 27.9, 27.6, 26.3, 23.2, 22.6, 14.0, 13.7; ^{19}F NMR (282 MHz, $CDCl_3$) δ -81.5 (t, *J* = 9.9 Hz, 3 F), -110.8 (t, *J* = 11.0 Hz, 2 F), -124.3~-124.5 (m, 2 F), -126.8~-127.0 (m, 2 F); IR (neat) ν (cm^{-1}) 2962, 2930, 2859, 1700, 1625, 1563, 1468, 1403, 1351, 1235, 1135, 1092, 1048; MS (70 ev, EI) *m/z* (%) 426 (M^+, 26.24), 257 (100); HRMS calcd for $C_{18}H_{23}F_9O$ (M^+): 426.1605, Found: 426.1602.

3. Typical Procedure III: Hydrolytic defluorination reaction of fully substituted furans **2a-2e**.

(1) Synthesis of (4,5-diphenyl-3-(*n*-propyl)furan-2-yl) *n*-perfluoropropyl ketone (**3a**)

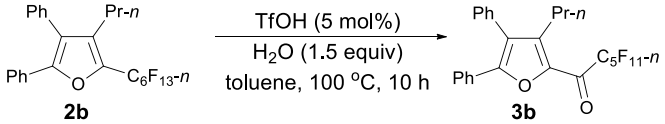


2a **3a** 97%

Typical Procedure III: To a dried Schlenk tube were added furan **2a** (0.0962 g, 0.20 mmol)/rinsed with toluene (2 mL), TfOH (0.9 μL, *d* = 1.695 g/mL, 0.0015 g, 0.01 mmol), and H_2O (5.4 μL, *d* = 1.0 g/mL, 0.0054 g, 0.3 mmol) at room temperature under a nitrogen atmosphere. The resulting reaction mixture was placed in a pre-heated oil bath of 100 °C and stirried for 12 h as monitored by TLC. After being cooled to the room temperature, the crude reaction mixture was transferred to a round bottom flask with Et_2O (3 × 10 mL). After removing the solvent via rotary evaporation, column chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 100/1) afforded **3a**[14] (0.0889 g, 97%) as a solid: m.p. 54.4-56.5 °C (*n*-hexane/ethyl acetate); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.53-7.43 (m, 5 H, ArH), 7.35-7.24 (m, 5 H, ArH), 2.76-2.63 (m, 2 H, CH_2), 1.57-1.41 (m, 2 H, CH_2), 0.86 (t, *J* = 7.4 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.5 (t, *J* = 25.8 Hz), 154.5, 145.9, 143.4, 131.4, 130.0, 129.7, 129.2, 128.9, 128.7, 128.5, 127.0, 126.8, 26.6, 22.6, 14.0; ^{19}F NMR (282 MHz, $CDCl_3$) δ -80.9 (t, *J* = 9.6 Hz, 3 F), -117.7~-117.9 (m, 2 F), -126.1~-126.3 (m, 2 F).

The following compounds **3b-3e** were prepared following **Typical Procedure III**.

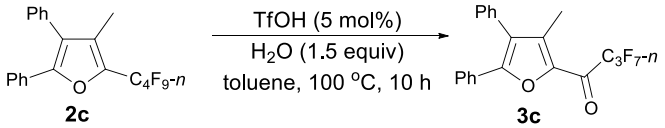
(2) Synthesis of (4,5-diphenyl-3-(*n*-propyl)furan-2-yl) *n*-perfluoropentyl ketone (**3b**)



2b **3b** 95%

The reaction of furan **2b** (0.1162 g, 0.20 mmol) with TfOH (99%, 0.9 μL, *d* = 1.695 g/mL, 0.0015 g, 0.01 mmol), and H_2O (5.4 μL, *d* = 1.0 g/mL, 0.0054 g, 0.3 mmol) in toluene (2 mL) at 100 °C for 10 h afforded **3b**[14] (0.1059 g, 95%) (petroleum ether/ethyl acetate = 100/1) as an oil: ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.53-7.42 (m, 5 H, ArH), 7.36-7.25 (m, 5 H, ArH), 2.76-2.62 (m, 2 H, CH_2), 1.57-1.41 (m, 2 H, CH_2), 0.86 (t, *J* = 7.5 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.5 (t, *J* = 26.1 Hz), 154.5, 146.0, 143.4, 131.4, 130.0, 129.7, 129.2, 128.9, 128.7, 128.5, 127.0, 126.8, 26.6, 22.6, 14.0; ^{19}F NMR (282 MHz, $CDCl_3$) δ -81.2 (tt, J_1 = 9.7 Hz, J_2 = 2.4 Hz, 3 F), -116.8~-117.1 (m, 2 F), -121.7~-122.1 (m, 2 F), -122.6~-123.0 (m, 2 F), -126.4~-126.8 (m, 2 F).

(3) Synthesis of (4,5-diphenyl-3-methylfuran-2-yl) *n*-perfluoropropyl ketone (**3c**)



2c **3c** 95%

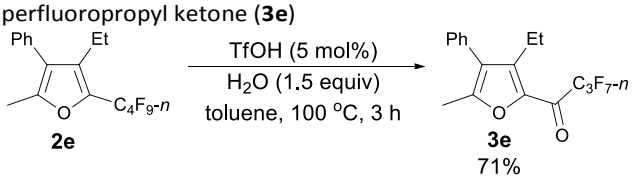
The reaction of furan **2c** (0.0903 g, 0.20 mmol) with TfOH (99%, 0.9 μL, *d* = 1.695 g/mL, 0.0015 g, 0.01 mmol), and H_2O (5.4 μL, *d* = 1.0 g/mL, 0.0054 g, 0.3 mmol) in toluene (2 mL) at 100 °C for 10 h afforded **3c**[14] (0.0812 g, 95%) (petroleum ether/ethyl acetate = 100/1) as a solid: m.p. 55.3-56.7 °C (*n*-hexane/ethyl acetate); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.55-7.44 (m, 5 H, ArH), 7.37-7.25 (m, 5 H, ArH), 2.33 (s, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.9 (t, *J* = 26.0 Hz), 154.5, 143.6, 141.2, 131.2, 129.9, 129.8, 129.2, 128.9, 128.7, 128.5, 127.2, 126.9, 10.9; ^{19}F NMR (282 MHz, $CDCl_3$) δ -80.9 (t, *J* = 9.0 Hz, 3 F), -117.8~-118.1 (m, 2 F), -126.1~-126.3 (m, 2 F).

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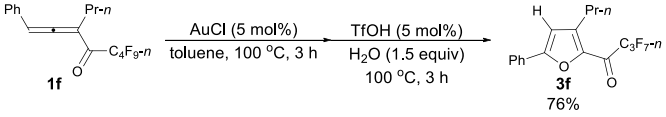
(4) Synthesis of (3-ethyl-5-methyl-4-phenylfuran-2-yl) *n*-perfluoropropyl ketone (**3e**)



The reaction of furan **2e** (0.0809 g, 0.20 mmol) with TfOH (99%, 0.9 μL, *d* = 1.695 g/mL, 0.0015 g, 0.01 mmol), and H_2O (5.4 μL, *d* = 1.0 g/mL, 0.0015 g, 0.3 mmol) in toluene (2 mL) at 100 °C for 3 h afforded **3e**[14] (0.0543 g, 71%) (petroleum ether/ethyl acetate = 100/1) as an oil: ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.51-7.37 (m, 3 H, ArH), 7.28-7.22 (m, 2 H, ArH), 2.78 (q, *J* = 7.5 Hz, 2 H, CH_2), 2.38 (s, 3 H, CH_3), 1.10 (t, *J* = 7.5 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.1 (t, *J* = 26.0 Hz), 156.9, 146.2, 143.4, 130.8, 129.5, 128.8, 128.0, 127.1, 18.3, 13.5, 13.0; ^{19}F NMR (282 MHz, $CDCl_3$) δ -81.0 (t, *J* = 8.9 Hz, 3 F), -117.8~-117.9 (m, 2 F), -126.3~-126.4 (m, 2 F).

4. Typical Procedure IV: The two-step one-pot tandem cycloisomerization/hydrolytic defluorination reaction of 3-monoaryl-substituted allenyl perfluorobutyl ketones **1f-1k**.

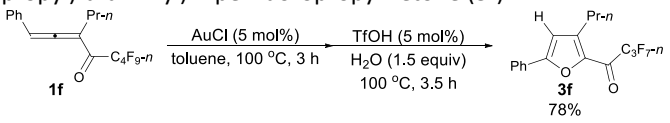
(1) Synthesis of (5-phenyl-3-(*n*-propyl)furan-2-yl) *n*-perfluoropropyl ketone (**3f**)



Typical Procedure IV: To a dried Schlenk tube were added anhydrous AuCl (97%, 0.0121 g, 0.051 mmol), 1-phenylhexa-1,2-dien-3-yl *n*-perfluorobutyl ketone **1f** (0.4035 g, 1.00 mmol), and anhydrous toluene (5 mL) at room temperature under N_2 atmosphere. The resulting mixture was then placed in a pre-heated oil bath of 100 °C and stirred. After 3 h as monitored by TLC the cycloisomerization reaction was over, TfOH (99%, 4.5 μL, *d* = 1.695 g/mL, 0.0076 g, 0.050 mmol) and H_2O (27 μL, *d* = 1.0 g/mL, 0.0270 g, 1.50 mmol) were added to the above crude reaction mixture, and the resulting mixture was stirred at 100 °C for another 3 h as monitored by TLC. After being cooled to the room temperature, the reaction mixture was transferred to a round bottom flask. After removing the solvent via rotary evaporation, column chromatography on silica gel (eluent: petroleum ether, then petroleum ether/ethyl acetate = 500/1) afforded **3f**[14] (0.2910 g, 76%) as a liquid: ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.84-7.71 (m, 2 H, ArH), 7.51-7.36 (m, 3 H, ArH), 6.82 (s, 1 H, =CH), 2.90 (t, *J* = 7.7 Hz, 2 H, CH_2), 1.80-1.59 (m, 2 H, CH_2), 1.01 (t, *J* = 7.5 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.4 (t, *J* = 25.8 Hz), 159.1, 146.9, 143.9, 130.3, 129.1, 128.5, 125.4, 110.4, 28.4, 22.3, 13.8 (d, *J* = 2.1 Hz); ^{19}F NMR (282 MHz, $CDCl_3$) δ -80.9~-81.1 (m, 3 F), -118.0 (q, *J* = 8.8 Hz, 2 F), -126.3 (s, 2 F).

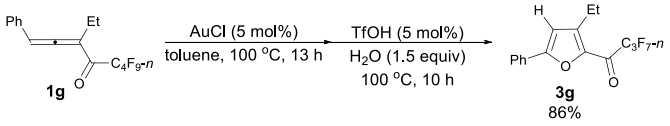
The following compounds **3f-3k** were prepared following **Typical Procedure IV**.

(2) Gram scale reaction for the synthesis of (5-phenyl-3-(*n*-propyl)furan-2-yl) *n*-perfluoropropyl ketone (**3f**)



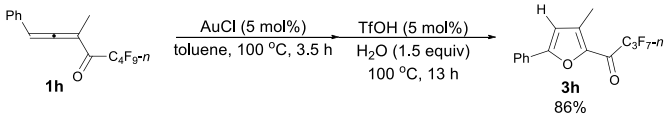
The reaction of 1-phenylhexa-1,2-dien-3-yl *n*-perfluorobutyl ketone **1f** (1.0102 g, 2.50 mmol) with AuCl (97%, 0.0301 g, 0.126 mmol) in toluene (12.5 mL) at 100 °C for 3 h afforded crude product **2f**. Then TfOH (99%, 11.2 μL, *d* = 1.695 g/mL, 0.0188 g, 0.125 mmol) and H_2O (67.5 μL, *d* = 1.0 g/mL, 0.0675 g, 3.75 mmol) were added and reacted at 100 °C for another 3.5 h to afford **3f**[14] (0.7448 g, 78%) (petroleum ether/ethyl acetate = 500/1) as an oil: ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.83-7.74 (m, 2 H, ArH), 7.54-7.39 (m, 3 H, ArH), 6.83 (s, 1 H, =CH), 2.91 (t, *J* = 7.7 Hz, 2 H, CH_2), 1.80-1.63 (m, 2 H, CH_2), 1.02 (t, *J* = 7.4 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.4 (t, *J* = 25.9 Hz), 159.0, 146.8, 143.9, 130.2, 129.1, 128.5, 125.4, 110.3, 28.4, 22.3, 13.8; ^{19}F NMR (282 MHz, $CDCl_3$) δ -81.0 (t, *J* = 8.9 Hz, 3 F), -118.0 (q, *J* = 9.5 Hz, 2 F), -126.3 (s, 2 F).

(3) Synthesis of (3-ethyl-5-phenylfuran-2-yl) *n*-perfluoropropyl ketone (**3g**)



The reaction of 1-phenylpenta-1,2-dien-3-yl perfluorobutyl ketone **1g** (0.1945 g, 0.50 mmol) with AuCl (97%, 0.0061 g, 0.026 mmol) in toluene (5 mL) at 100 °C for 13 h afforded crude product **2g**. Then TfOH (99%, 2.3 μL, *d* = 1.695 g/mL, 0.0039 g, 0.026 mmol) and H_2O (13.5 μL, *d* = 1.0 g/mL, 0.0135 g, 0.75 mmol) were added and reacted at 100 °C for another 10 h to afford **3g**[14] (0.1581 g, 86%) (petroleum ether/ethyl acetate = 500/1) as a solid: m.p. 40.2-41.2 °C (*n*-hexane/ethyl acetate); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.89-7.69 (m, 2 H, ArH), 7.59-7.33 (m, 3 H, ArH), 6.86 (s, 1 H, =CH), 2.96 (q, *J* = 7.4 Hz, 2 H, CH_2), 1.29 (t, *J* = 7.5 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.3 (t, *J* = 26.3 Hz), 159.2, 148.3, 143.6, 130.3, 129.1, 128.5, 125.4, 109.8, 20.0, 13.1; ^{19}F NMR (282 MHz, $CDCl_3$) δ -81.0 (t, *J* = 8.9 Hz, 3 F), -118.1 (q, *J* = 9.0 Hz, 2 F), -126.3 (s, 2 F).

(4) Synthesis of (3-methyl-5-phenylfuran-2-yl) *n*-perfluoropropyl ketone (**3h**)



The reaction of 1-phenylbuta-1,2-dien-3-yl perfluorobutyl ketone **1h** (0.3763 g, 1.00 mmol) with AuCl (97%, 0.0111 g, 0.046 mmol) in toluene (5 mL) at 100 °C for 3.5 h afforded crude product **2h**. Then TfOH (99%, 4.5 μL, *d* = 1.695 g/mL, 0.0076 g, 0.050 mmol) and H_2O (27 μL, *d* = 1.0 g/mL, 0.0270 g, 1.50 mmol) were added and reacted at 100 °C for another 13 h to afford **3h**[14] (0.3039 g, 86%) (petroleum ether/ethyl acetate = 500/1) as an oil: ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.82-7.70 (m, 2

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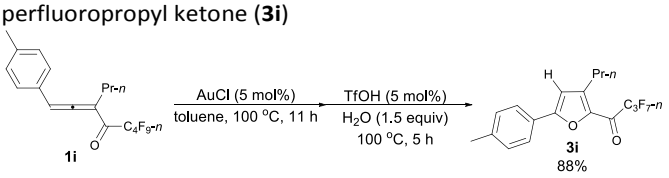
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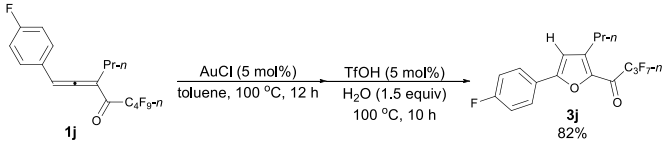
H, ArH), 7.53-7.38 (m, 3 H, ArH), 6.80 (s, 1 H, =CH), 2.52 (s, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.5 (t, *J* = 26.2 Hz), 158.9, 144.2, 141.9, 130.3, 129.1, 128.4, 125.4, 111.6, 12.6; ^{19}F NMR (282 MHz, $CDCl_3$) δ -80.5 (t, *J* = 9.2 Hz, 3 F), -117.5~-117.9 (m, 2 F), -125.8~-126.0 (m, 2 F).

(5) Synthesis of (5-(4'-methylphenyl)-3-(*n*-propyl)furan-2-yl) *n*-perfluoropropyl ketone (**3i**)



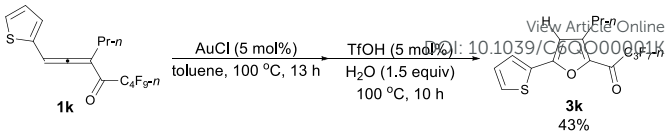
The reaction of 1-(4'-methylphenyl)hexa-1,2-dien-3-yl perfluorobutyl ketone **1i** (0.2101 g, 0.50 mmol) with AuCl (97%, 0.0058 g, 0.024 mmol) in toluene (5 mL) at 100 °C for 11 h afforded crude product **2i**. Then TfOH (99%, 2.3 μL, *d* = 1.695 g/mL, 0.0039 g, 0.026 mmol) and H_2O (13.5 μL, *d* = 1.00 g/mL, 0.0135 g, 0.75 mmol) were added and reacted at 100 °C for another 5 h to afford **3i**[14] (0.1756 g, 88%) (petroleum ether/ethyl acetate = 500/1) as a solid: m.p. 33.5-34.0 °C (*n*-hexane/ethyl acetate); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.67 (d, *J* = 7.8 Hz, 2 H, ArH), 7.27 (d, *J* = 7.8 Hz, 2 H, ArH), 6.78 (s, 1 H, =CH), 2.90 (t, *J* = 7.5 Hz, 2 H, CH_2), 2.41 (s, 3 H, CH_3), 1.77-1.63 (m, 2 H, CH_2), 1.01 (t, *J* = 7.4 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.1 (t, *J* = 25.9 Hz), 159.4, 147.0, 143.7, 140.7, 129.8, 125.8, 125.5, 109.8, 28.4, 22.3, 21.5, 13.8; ^{19}F NMR (282 MHz, $CDCl_3$) δ -81.0 (t, *J* = 8.3 Hz, 3 F), -118.0 (q, *J* = 8.7 Hz, 2 F), -126.4 (s, 2 F).

(6) Synthesis of (5-(4'-fluorophenyl)-3-(*n*-propyl)furan-2-yl) *n*-perfluoropropyl ketone (**3j**)



The reaction of 1-(4'-fluorophenyl)hexa-1,2-dien-3-yl perfluorobutyl ketone **1j** (0.2112 g, 0.50 mmol) with AuCl (97%, 0.0061 g, 0.026 mmol) in toluene (5 mL) at 100 °C for 12 h afforded crude product **2j**. Then TfOH (99%, 2.3 μL, *d* = 1.695 g/mL, 0.0039 g, 0.026 mmol) and H_2O (13.5 μL, *d* = 1.0 g/mL, 0.0135 g, 0.75 mmol) were added and reacted at 100 °C for another 10 h to afford **3j**[14] (0.1648 g, 82%) (petroleum ether/ethyl acetate = 500/1) as an oil: ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.77 (dd, J_1 = 8.3 Hz, J_2 = 5.6 Hz, 2 H, ArH), 7.17 (t, *J* = 8.4 Hz, 2 H, ArH), 6.78 (s, 1 H, =CH), 2.90 (t, *J* = 7.7 Hz, 2 H, CH_2), 1.78-1.62 (m, 2 H, CH_2), 1.02 (t, *J* = 7.4 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.3 (t, *J* = 25.7 Hz), 163.8 (d, *J* = 250.4 Hz), 158.0, 146.9, 143.9, 127.5 (d, *J* = 7.7 Hz), 124.9 (d, *J* = 3.8 Hz), 116.4 (t, *J* = 22.2 Hz), 110.1, 28.4, 22.3, 13.8; ^{19}F NMR (282 MHz, $CDCl_3$) δ 81.0 (t, *J* = 8.9 Hz, 3 F), -109.3~-109.6 (m, 1 F), -118.0 (q, *J* = 8.9 Hz, 2 F), -126.4 (s, 2 F).

(7) Synthesis of (3-(*n*-propyl)-5-(2'-thienyl)furan-2-yl) *n*-perfluoropropyl ketone (**3k**)



The reaction of 1-(2'-thienyl)hexa-1,2-dien-3-yl perfluorobutyl ketone **1k** (0.0812 g, 0.20 mmol) with AuCl (97%, 0.0025 g, 0.010 mmol) in toluene (2 mL) at 100 °C for 13 h afforded crude product **2k**. Then TfOH (99%, 0.9 μL, *d* = 1.695 g/mL, 0.0015 g, 0.010 mmol) and H_2O (5.4 μL, *d* = 1.0 g/mL, 0.0054 g, 0.30 mmol) were added and reacted at 100 °C for another 10 h to afford **3k**[14] (0.0331 g, 43%) (petroleum ether/ethyl acetate = 500/1) as an oil: ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.52 (d, *J* = 3.6 Hz, 1 H, ArH), 7.45 (d, *J* = 4.8 Hz, 1 H, ArH), 7.13 (d, *J* = 4.4 Hz, 1 H, ArH), 6.67 (s, 1 H, =CH), 2.89 (t, *J* = 7.5 Hz, 2 H, CH_2), 1.79-1.60 (m, 2 H, CH_2), 1.01 (t, *J* = 7.4 Hz, 3 H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.0 (t, *J* = 26.4 Hz), 154.4, 147.0, 143.3, 131.4, 128.38, 128.36, 127.0, 110.0, 28.4, 22.2, 13.8; ^{19}F NMR (282 MHz, $CDCl_3$) δ -81.0 (t, *J* = 9.0 Hz, 3 F), -118.0 (q, *J* = 8.8 Hz, 2 F), -126.3~-126.4 (m, 2 F).

Acknowledgements

Financial support from the National Natural Science Foundation of China (21572202) is greatly appreciated. Shengming Ma is a Qiu Shi Adjunct Professor at Zhejiang University. The results presented in entry 2 of Table 2, entry 5 of Table 4 and entry 1 of Table 8 have been kindly reproduced by Mr. Wangteng Wu and Dr. Jiasheng Zhang, respectively.

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